ATOM COUNT CARD

6% HYDROGEN PEROXIDE: FROM A PRINTED LABEL TO MOLES, MOLECULES AND ATOMS

MOLE CONCEPT ACTIVITY 2



1. PRODUCT IDENTITY

- **Product:** Hydrogen Peroxide
- **Compound:** H_2O_2
- **Why it matters:** it is one of the most versatile, environmentally friendly, and powerful oxidizing agents on Earth.
- **Student:** Jemar B. Plotado

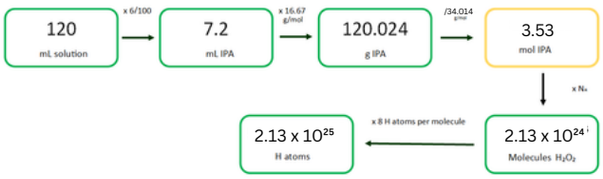
2. BASE QUANTITATIVE DATA

QUANTITY	VALUE	SOURCE	PRECISION/SIG.FIG.
Net Content	120 mL	Product Label	Nominal; not limiting
Concentration	6% solution	Product Label	1 s.f.
Density, Pure IPA	16.67 g/mL	Product Label	4 s.f.
Atomic masses H, O	1.008, 15.999	Ptable (n.d.)	Not limiting
Avogadro Constant	$N_A = 6.02214076 \times 10^{23}$	TranslatorCafe.com (n.d.)	Exact
Formula H_2O_2	Subscripts 3,8,1	Ptable (n.d.)	Exact counts

3. PRODUCT IDENTITY

ELEMENT	NUMBER OF ATOMS	ATOMIC MASS (g/mol each)	CONTRIBUTION (g/mol)	% by mass
H	2	1.008	2.016	5.926%
O	2	15.99	31.998	94.074%
H_2O_2	4	_	34.014 g/mol	100.000%

4. THE WHOLE MAP: 120 mL bottle



5. MOLE-CONVERSION CALCULATIONS

1. MOLAR MASS OF H_2O_2

$$2(1.008) + 2(15.999) = 34.014 \text{ g/mol}$$

2. Label bridge: volume → mass of H_2O_2

$$120 \text{ mL solution} \times \frac{6 \text{ g } H_2O_2}{100 \text{ mL solution}} = 7.2 \text{ g } H_2O_2$$

$$= \boxed{7.2 \text{ g } H_2O_2}$$

3. Mass → moles

$$7.2 \text{ g } H_2O_2 \times \frac{1 \text{ mol } H_2O_2}{34.014 \text{ g } H_2O_2} = 0.211675 \text{ mol}$$

$$\Rightarrow \boxed{0.212 \text{ mol } H_2O_2}$$

4. Moles → molecules

$$0.211675 \text{ mol } H_2O_2 \times \frac{6.02214076 \times 10^{23} \text{ molecules } H_2O_2}{1 \text{ mol } H_2O_2} = 1.2746 \times 10^{23} \text{ molecules}$$

$$\Rightarrow \boxed{1.27 \times 10^{23} \text{ molecules}}$$

5. Atoms of one element inside the formula

$$1.2746 \times 10^{23} \text{ molecules } H_2O_2 \times \frac{2 \text{ O atoms}}{1 \text{ molecule } H_2O_2} = 2.5492 \times 10^{23} \text{ O atoms}$$

$$\Rightarrow \boxed{2.55 \times 10^{23} \text{ O atoms}}$$

6. Percent composition + two-route check (oxygen)

$$\%O = \frac{31.998 \text{ g/mol}}{34.014 \text{ g/mol}} \times 100 = \boxed{94.074\%}$$

Route A: Use the oxygen mass fraction

$$7.2 \text{ g } H_2O_2 \times 0.94074 = \boxed{6.77 \text{ g O}}$$

Route B: Convert through moles

$$7.2 \text{ g } H_2O_2 \times \frac{1 \text{ mol } H_2O_2}{34.014 \text{ g } H_2O_2} \times \frac{2 \text{ mol O}}{1 \text{ mol } H_2O_2} \times \frac{15.999 \text{ g O}}{1 \text{ mol O}} = 6.77 \text{ g O}$$

7. Reverse: from % composition to the formula

$$H = 5.926\% \left(\frac{100 \text{ g}}{100\%}\right) = 5.926 \text{ g} \left(\frac{1 \text{ mol}}{1.008 \text{ g}}\right) = 5.879 \text{ mol}$$

$$O = 94.074\% \left(\frac{100 \text{ g}}{100\%}\right) = 94.074 \text{ g} \left(\frac{1 \text{ mol}}{15.999 \text{ g}}\right) = 5.880 \text{ mol}$$

$$H = \frac{5.879}{5.879} = 1 \qquad O = \frac{5.880}{5.879} \approx 1$$

Empirical formula = HO

$$M_{HO} = 1.008 + 15.999 = 17.007 \text{ g/mol}$$

$$n = \frac{34.014}{17.007} = 2$$

Molecular formula = $(HO)_2 = \boxed{H_2O_2}$

8. Reverse: how much solution contains one mole?

$$1.00 \text{ mol } H_2O_2 \times \frac{34.014 \text{ g } H_2O_2}{1 \text{ mol } H_2O_2} \times \frac{100 \text{ mL solution}}{6 \text{ g } H_2O_2} = 566.9 \text{ mL solution}$$

6. SCALE-UP: FROM ONE BOTTLE TO THE WHOLE CAMPUS

1.) BOTTLE

120 mL of 6% solution

0.212 mol

1.7 x 10^{23} molecules

2.) ONE LAB ONE YEAR

assume 20 bottles per year

5.08 mol

5.08 mol x 34.014 g/mol = 172.8 g H_2O_2

3.) CAMPUS, ONE YEAR

assume 5 such labs

50.8 mol

50.8 mol x 34.014 g/mol = 1.73 kg H_2O_2

4.) REVERSE

ane mole of IPA is in

567 mL

About 4.7 bottles of 120 mL.

7. SIGNIFICANT-FIGURE CHECK

CALCULATOR: 2.125815688x10^{24} molecules
REPORTED: 2.1x10^{24} molecules

- **LIMITS:** 6% (1 s.f.). Exact: NAA, subscripts, bottle counts.
- "120 mL" (2 s.f.) EXACT
- Guard digits kept until last line of every calculation.

AI THAT I USE

CHATGPT AND GEMINI: I used this AI for a exact measurements and checking for my personal formulation and calculations.